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Levi M. Barker Ph.D.

Education

Ph.D. in Mechanical Engineering Spring 2027
University of Utah | Salt Lake City, Utah

- Magnetics & Medical Robotics under Dr. Jake J. Abbott
- Dissertation Title: "Linear Modeling and Model-based Manipulation of Conductive Nonmagnetic Objects Using Magnetic Induction"

B.S./M.S. in Mechanical Engineering Spring 2025
University of Utah | Salt Lake City, Utah

- Realized a novel stroke rehabilitation prototype for remapping neural pathways (Pat. Pend.)
- Lead a mechatronics team in design and creation of a confection sorting robot

Publications

1. **L.M. Barker**, T.J. Allen, J.J. Abbott, "Force and Torque Induced on a Conductive Nonmagnetic Cylinder by a Rotating Magnetic Dipole." *IEEE Transactions on Magnetics*, (under review)
2. T.J. Allen, **L.M. Barker**, J.J. Abbott, "Assume a Spherical Cube: Bounding the Force and Torque Induced on a Conductive Nonmagnetic Cube by a Rotating Magnetic Dipole." *IEEE Magnetics Letters*, 2026
3. **L.M. Barker**, T.J. Allen, A.J. Sperry, J.J. Abbott, "Revisiting the far-field model of the force and torque induced on a conductive nonmagnetic sphere by a rotating magnetic dipole." *Scientific Reports*, 2025

Relevant Experience

Instructor: ME EN 3220 Dynamic Systems and Control Summer 2026
University of Utah | Salt Lake City, Utah

- Taught and graded 20 undergraduate students
- Digitized and converted department content to WCAG AA 2.1 accessibility standards

Engineering Tutor Dec. 2021 – May 2023
University of Utah | Salt Lake City, Utah

- Tutored 50 students weekly in mathematics, programming, statics, dynamics, robotics, manufacturing, design and CAD modeling
- Encouraged critical thinking, time management, and test taking skills
- Facilitated a safe and welcoming environment for learning

Undergraduate Researcher Summer 2022
Human Centered Haptics and Robotics Lab | Salt Lake City, Utah

- Characterized materials for utilization within robotic prosthetics and force sensing devices
- Designed, manufactured, and programmed a custom linear indentation instrument
- Proposed and secured an Undergraduate Research Opportunity Program grant

Continuous Improvement Intern Summer 2021
Maersk | Salt Lake City, Utah

- Implemented aspects of Lean, Six Sigma, and 5S in a warehouse-within-a-warehouse
- Analyzed and improved productivity by introducing protocol and personnel training reducing annual overhead by $40,000

Pyrolysis Plant Operator Aug 2019 – May 2020
USTAR Bioenergy Center | Logan, Utah

- Collected and analyzed production samples, in accordance with safety protocol
- Troubleshooted and rectified mechanical, electrical, and thermodynamical problems

Recognition

- **Accessibility Training Certificate** – Price College of Engineering 2026
- **Student Leader of the Year** – Department of Mechanical Engineering 2023
- **Certified Solidworks Associate** – Dassault Systemes 2018

Skills

Technical skills: Design of experiments, data visualization, CAD modeling, PCB design
Programming: OOP, `Python`, `LaTeX`, `Git`, `MATLAB`, `Simulink`, `Arduino`
Soft skills: Lifelong learner, critical thinker, team leader, detail oriented